WORKED SOLUTION

STEP II 2010, Question 13

Mechanics • Elementary probability

STEP II 2010, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q13

The key idea

Rosalind passes exactly when she wins at least one game against Pardeep *and* at least one against Quentin. "At least one" is always easiest through the complement: the chance of winning at least one of n games is 1 minus the chance of losing them all. With independent games, each strategy's success probability is then a product of two such factors, and comparing strategies means factorising a difference. Writing $a = 1 - p$ and $b = 1 - q$ for the losing probabilities keeps the algebra tidy.

Solution

We assume that the results of different games are independent, and that each game against Pardeep (or Quentin) is won with probability p (or q) whatever has happened before. Order doesn't matter: whether or not play stops once she has qualified, she qualifies in exactly the same outcomes.

If Rosalind plays Pardeep m times and Quentin n times, let A be the event that she wins at least one game against Pardeep and B the event that she wins at least one against Quentin. Then $\mathrm{P}(A) = 1 - (1-p)^m$, since A fails only if she loses all m games, and similarly $\mathrm{P}(B) = 1 - (1-q)^n$. The two events involve different games, so they are independent, and

$$\mathrm{P}(\text{accepted}) = (1 - (1-p)^m)(1 - (1-q)^n).$$

Throughout, write $a = 1 - p$ and $b = 1 - q$. Since $0 < p < q < 1$, we have $0 < b < a < 1$.

Part (i)

Playing Pardeep twice and Quentin once gives success probability $(1 - a^2)(1 - b)$; playing Quentin twice and Pardeep once gives $(1 - a)(1 - b^2)$. Factorising each difference

of squares,

$$\begin{aligned}(1-a^2)(1-b)-(1-a)(1-b^2) &= (1-a)(1-b)[(1+a)-(1+b)]\\ &= (1-a)(1-b)(a-b)\\ &= pq(q-p).\end{aligned}$$

This is strictly positive because $p, q > 0$ and $q > p$. So playing Pardeep twice gives the larger chance of acceptance, and that is what she should choose.

Part (ii)

The three strategies succeed with probabilities

$$S_1 = (1-a^3)(1-b), \qquad S_2 = (1-a^2)(1-b^2), \qquad S_3 = (1-a)(1-b^3).$$

Each contains the factor $(1-a)(1-b) = pq$, because $1-x^k = (1-x)(1+x+\cdots+x^{k-1})$.

Strategy 1 always beats strategy 3. We have

$$\begin{aligned}S_1 - S_3 &= pq[(1+a+a^2)-(1+b+b^2)]\\ &= pq\,(a-b)(1+a+b) > 0,\end{aligned}$$

since $a > b$. So strategy 3 is never the best choice, and only strategies 1 and 2 need comparing.

Strategy 1 against strategy 2. We have

$$S_1 - S_2 = pq[(1+a+a^2)-(1+a)(1+b)] = pq[a^2 - b(1+a)].$$

Returning to p and q,

$$S_1 - S_2 = pq\,D, \qquad \text{where} \qquad D = (1-p)^2 - (1-q)(2-p).$$

Since $pq > 0$, strategy 1 is better than strategy 2 exactly when $D > 0$.

Ask yourself How does the condition $q - p > \frac{1}{2}$ help to make D positive?

The only term in D involving q is $-(1-q)(2-p)$, and it increases as q increases, because $2 - p > 0$. So we replace q by the smallest value the condition allows. If $q - p > \frac{1}{2}$ then $1 - q < \frac{1}{2} - p$, and since $2 - p > 0$,

$$(1-q)(2-p) < (\tfrac{1}{2}-p)(2-p) = 1 - \tfrac{5}{2}p + p^2.$$

Therefore

$$D > (1-2p+p^2) - (1-\tfrac{5}{2}p+p^2) = \tfrac{1}{2}p > 0.$$

So $S_1 > S_2$, and together with $S_1 > S_3$ this shows that Rosalind should choose strategy 1 when $q - p > \frac{1}{2}$.

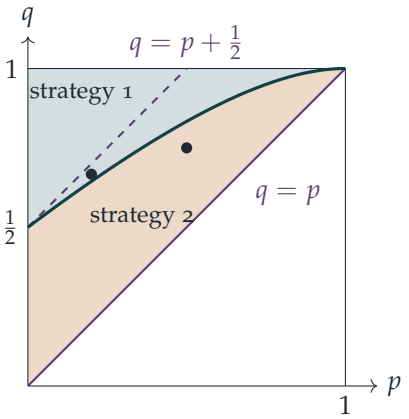
When $q - p < \frac{1}{2}$, either can win. Two examples:

- $p = \frac{1}{2}$, $q = \frac{3}{4}$ (so $q - p = \frac{1}{4}$). Here $D = \frac{1}{4} - \frac{1}{4} \cdot \frac{3}{2} = -\frac{1}{8} < 0$. Indeed $S_1 = \frac{7}{8} \cdot \frac{3}{4} = \frac{42}{64}$ and $S_2 = \frac{3}{4} \cdot \frac{15}{16} = \frac{45}{64}$, so **strategy 2 is better**.
- $p = \frac{1}{5}$, $q = \frac{2}{3}$ (so $q - p = \frac{7}{15} < \frac{1}{2}$). Here $D = \frac{16}{25} - \frac{1}{3} \cdot \frac{9}{5} = \frac{16}{25} - \frac{15}{25} = \frac{1}{25} > 0$. Indeed $S_1 = \frac{61}{125} \cdot \frac{2}{3} = \frac{122}{375}$ and $S_2 = \frac{9}{25} \cdot \frac{8}{9} = \frac{120}{375}$, so **strategy 1 is better**.

The whole picture is clear in the (p, q) plane. Setting $D = 0$ gives the boundary curve $q = 1 - \dfrac{(1-p)^2}{2-p}$. One can check that

$$\left(p + \tfrac{1}{2}\right) - \left(1 - \frac{(1-p)^2}{2-p}\right) = \frac{p}{2(2-p)} > 0,$$

so the curve lies just below the line $q = p + \frac{1}{2}$. Above the curve strategy 1 is best; between the curve and the diagonal $q = p$ strategy 2 is best.



The two dots are the examples above: $(\frac{1}{2}, \frac{3}{4})$ lies in the strategy 2 region and $(\frac{1}{5}, \frac{2}{3})$ just above the curve.

Checks

In part (i), let $q \to 1$: Quentin is a certain win, so only the Pardeep games matter and more of them is clearly better, which matches the difference $pq(q-p) \to p(1-p) > 0$. For part (ii) we enumerated all 2^4 outcomes exactly with fractions for 2000 random pairs $p < q$, confirming the formulas for S_1, S_2, S_3 and the factorisations. A simulation with $p = \frac{1}{2}$, $q = \frac{3}{4}$ gave $S_1 \approx 0.657$ and $S_2 \approx 0.703$, against the exact $\frac{21}{32} = 0.65625$ and $\frac{45}{64} \approx 0.7031$.

Common slips

- Computing "at least one win" as mp rather than $1 - (1-p)^m$; adding probabilities of overlapping events overcounts.

- Forgetting strategy 3: to show strategy 1 is best you must beat *both* alternatives.
- Giving examples with $q - p < \frac{1}{2}$ but not checking the condition $0 < p < q < 1$.

Beyond STEP

The intuition is "spend your effort where you are weakest", but the second part shows it isn't always right: the extra game against Pardeep has diminishing returns, and when p is not too small a second game against Quentin is worth more. Allocating a fixed budget of trials to maximise a product of "at least one success" probabilities is a small instance of the optimisation questions behind reliability theory and the design of experiments. Grimmett and Welsh, *Probability: An Introduction*, covers independence and complements carefully and has many problems in this spirit.

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